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SCHOOL OF INFORMATION AND COMMUNICATION

TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

TOPIC: Multi-Source Energy-Harvesting Autonomous Sensor Node.

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SYSTEM DESIGN

1. Introduction

This chapter presents the complete system design of the Multi-Source Energy-Harvesting Autonomous Sensor Node. The system is designed to operate autonomously using ambient energy sources while adapting its workload based on available energy to ensure implementation stability and reduce unnecessary complexity, the system adopts a simplified communication architecture. Instead of wireless or cloud-based (IoT) communication, telemetry data is transmitted directly via USB serial to a laptop for logging and analysis This approach ensures:

1. Minimal system complexity
2. Reduced firmware overhead
3. Deterministic communication
4. Stable experimental evaluation
5. Strong focus on adaptive energy management

2. System Objectives

The system is designed to:

1. Harvest energy from multiple ambient sources namely; solar, magnetic, and radio frequency.
2. Store harvested energy in super capacitors.
3. Monitor energy storage levels, adapt sensing and operational behavior based on energy availability, transmit telemetry data for evaluation by use of scheduled algorithms.

3. RESEARCH METHODOLOGY

Iterative Engineering Process Model

The research methodology model employs a design based and experimental approach, focusing on the development of a functional prototype and performance evaluation. The study follows an engineering design process consisting of system analysis, design, implementation, and testing.

Initially, a comprehensive literature review was conducted to understand existing energy harvesting technologies, power management strategies, and intelligent duty-cycling mechanisms. Based on the identified research gaps, system requirements are defined, followed by the design of the overall architecture and subsystem models. Hardware design is carried out using electronic design principles.

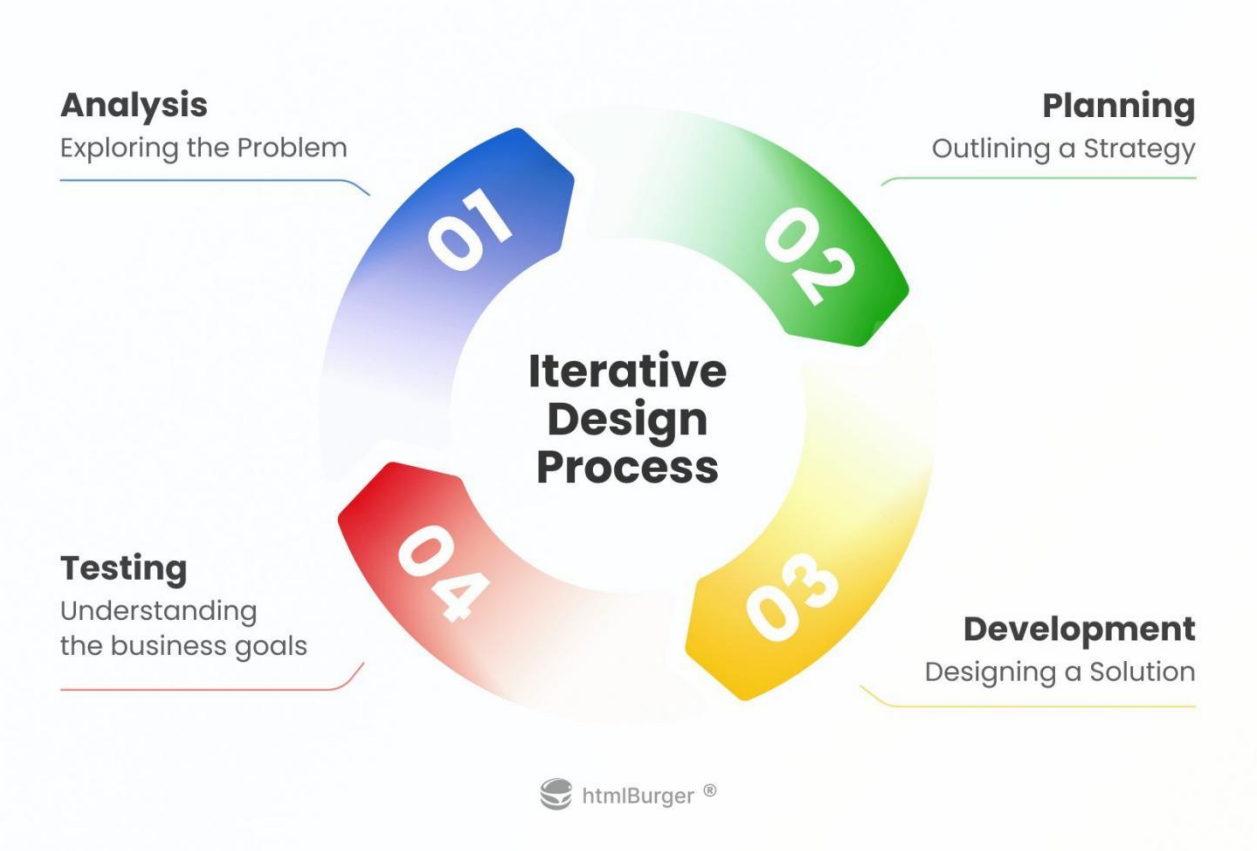


Figure 3: Iterative Engineering Process Model Diagram

METHODOLOGY STAGES

3.2.1 Stage One: Requirement Analysis

Extensive review of existing academic and industrial energy harvesting systems was conducted to identify design gaps. Functional and non-functional requirements were derived from this review.

Functional Requirements:

System Requirements

The proposed system shall:

1. harvest ambient energy from solar, magnetic, and radio-frequency sources.
2. convert and regulate harvested energy using a power management unit.
3. store harvested energy in a super capacitor-based storage unit.
4. monitor energy levels, estimate available energy, adaptive sensing and communication duty cycles according to the available energy.
5. transmit sensor data wirelessly using a low-power communication module.
6. enter low-power or sleep modes when available energy falls below the required operating threshold.

Non-Functional Requirements

The proposed system shall:

1. **Input energy from multiple ambient sources** (e.g., solar, magnetic, and RF harvesting modules) through integrated hardware interfaces designed to minimize overall power consumption.
2. **Process incoming harvested energy using power-management circuitry and embedded control algorithms** to regulate voltage levels and distribute power efficiently to system components.
3. **Output regulated electrical power to sensor nodes and communication modules**, ensuring stable and reliable operation.

3.2.2 Stage Two: System Architecture Design and Modeling

In the second stage, the overall system architecture of the proposed multi-source energy-harvesting autonomous sensor node is being designed. This stage is involving the definition of interactions among the energy harvesting units (solar, magnetic, and radio frequency), the power management subsystem, energy storage elements, the microcontroller unit, sensors, and the wireless communication module. System-level modeling is being performed to analyze energy flow, power conversion efficiency, and operational feasibility under varying ambient energy conditions. The architectural design is ensuring modularity to allow independent optimization of individual subsystems.

The system consists of four major subsystems:

1. Energy Harvesting Subsystem
2. Power Management and Storage Subsystem
3. Processing and Sensing Subsystem
4. Serial Communication and Data Logging Subsystem

3.2.2 Overview System Architecture

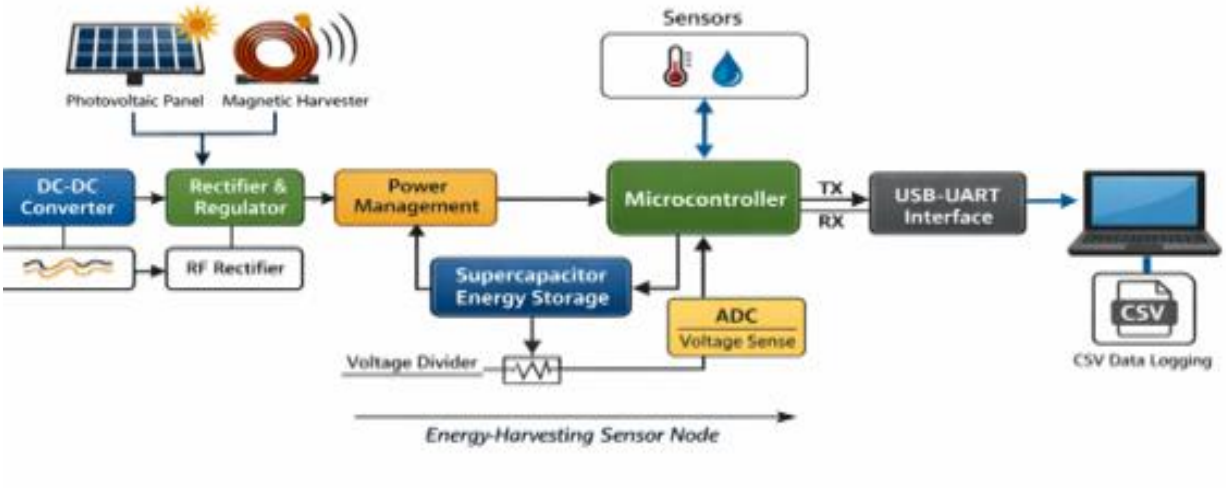


Figure 3.2.2: Overview High-Level System Block Diagram (a)

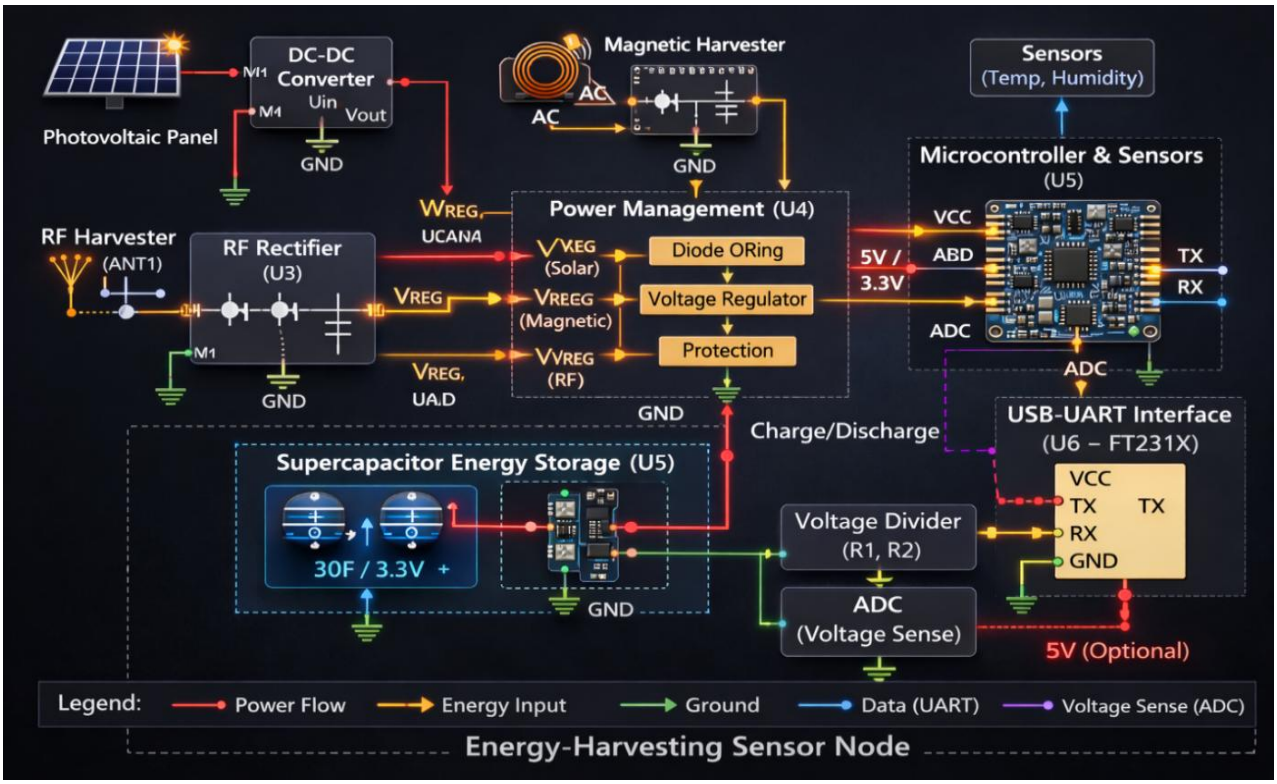


Figure 3.2.2: Detailed High-Level Circuit System Block Diagram (b)

3.2.3 Stage Three: Hardware Design and Component Selection

During this stage, detailed hardware design and component selection are being carried out. Electronic components are being selected based on ultra-low-power characteristics, efficiency, availability, and suitability for energy harvesting applications. Circuit schematics are being developed to integrate the energy harvesting sources, PMIC, super capacitor-based energy storage, microcontroller, sensors, and communication modules. Following schematic verification, the hardware design is being implemented through printed circuit board (PCB) design, ensuring compact layout, reduced power losses, and reliable electrical connectivity.

3.2.3.1 Electrical Design Specifications

To ensure a practical and implementable design, the following electrical parameters are defined:

Parameter	**Value**	**Justification**
Solar Panel Output	5V (nominal), 200 mA (max)	Suitable small PV module
Magnetic Harvester Output	2–8 V AC (variable)	Depends on field strength
RF Harvester Output	0.2–1.5 V DC (low power)	Ambient RF dependent
Regulated System Voltage	3.3 V ±5%	Standard MCU operating voltage
Super capacitor Rating	30 F, 3.0 V	Energy buffer
ADC Reference Voltage	3.3 V	Matches regulated supply
UART Baud Rate	9600 bps	Stable, low overhead

The regulated 3.3 V supply ensures compatibility with ultra-low-power microcontrollers and digital sensors.

3.2.3.2 Energy Harvesting Subsystem

The system integrates multiple ambient energy sources to increase reliability under varying environmental conditions**. (Gloomier Solar Power Station, 2021; Touwsrivier CPV Solar Project, 2025; Nzeanorue, 2025).**

3.2.3.2.1 Solar Harvesting Section

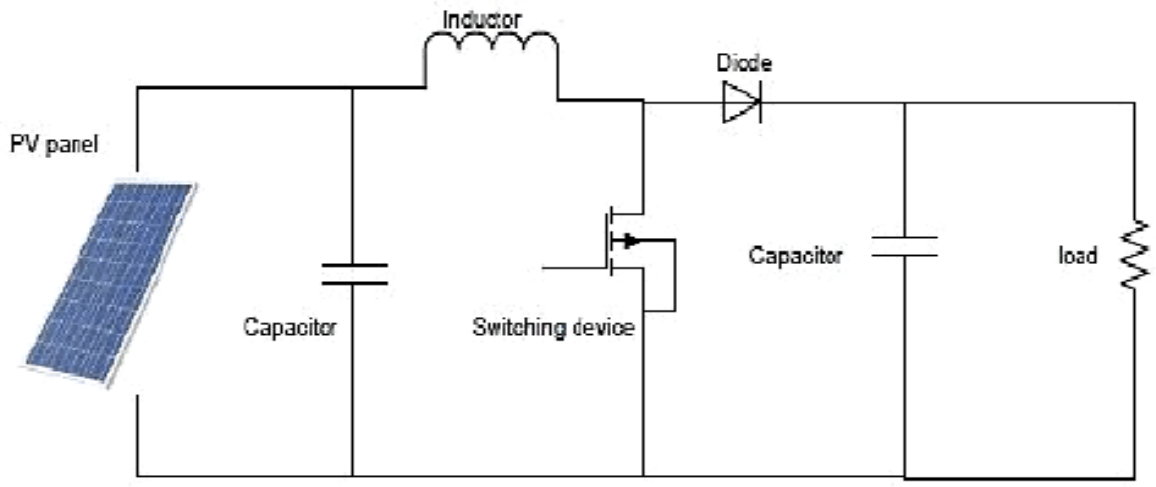


FIGURE 3.2.3.2.1: Solar Harvesting Section

Photovoltaic Panel

Converts sunlight into DC electrical energy. Provides variable voltage depending on light intensity.

DC-DC Converter

Steps up or regulates the solar voltage, ensures stable intermediate DC output and prevents under voltage operation of the system.

Role: Makes solar power usable for charging and regulation.

3.2.3.2.2 Magnetic Harvesting Section

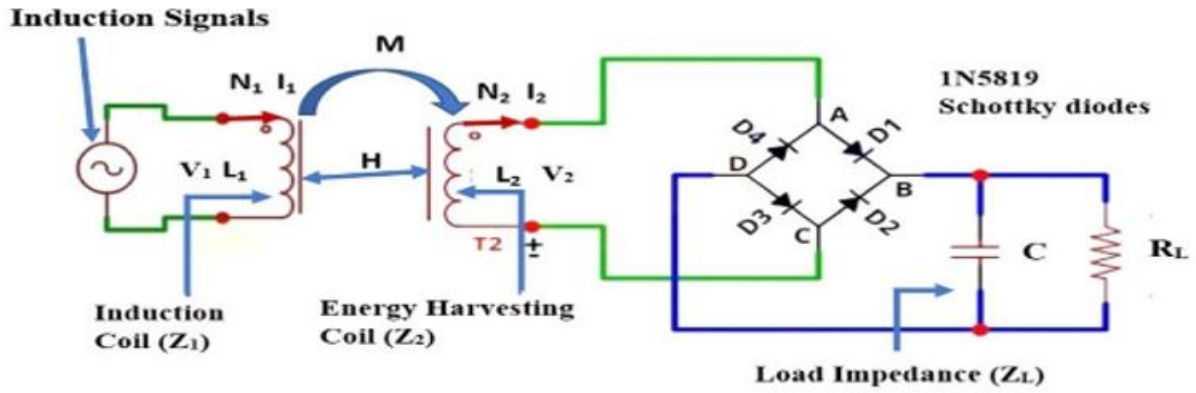


FIGURE 3.2.3.2.2: Magnetic Harvesting Section (a)

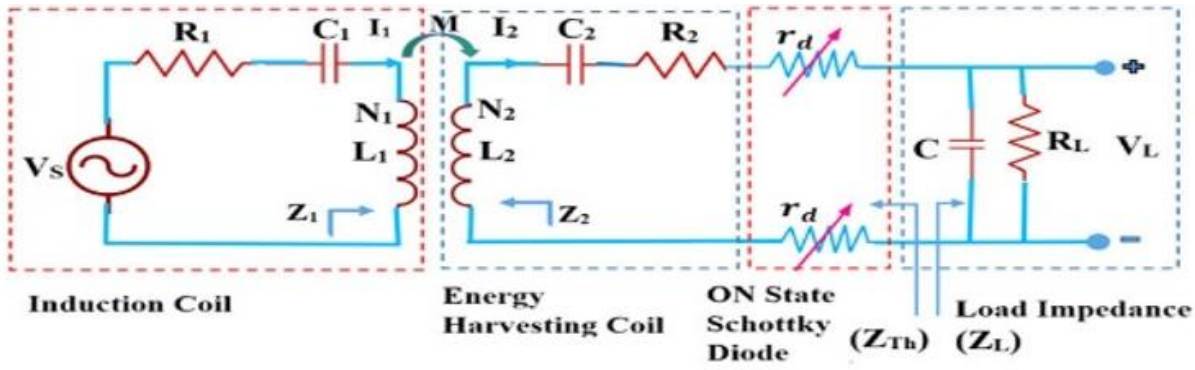


FIGURE 3.2.3.2.2: Magnetic Harvesting Section (b)

Magnetic Harvester (Coil + Core)

Converts changing magnetic fields into AC voltage (Faraday's Law), Output is typically low-level AC.

Rectifier & Filtering Network

Converts AC to DC, Smoothing ripple voltage.

Role: Converts ambient magnetic energy into usable DC power.

3.2.3.2.3 RF Harvesting Section

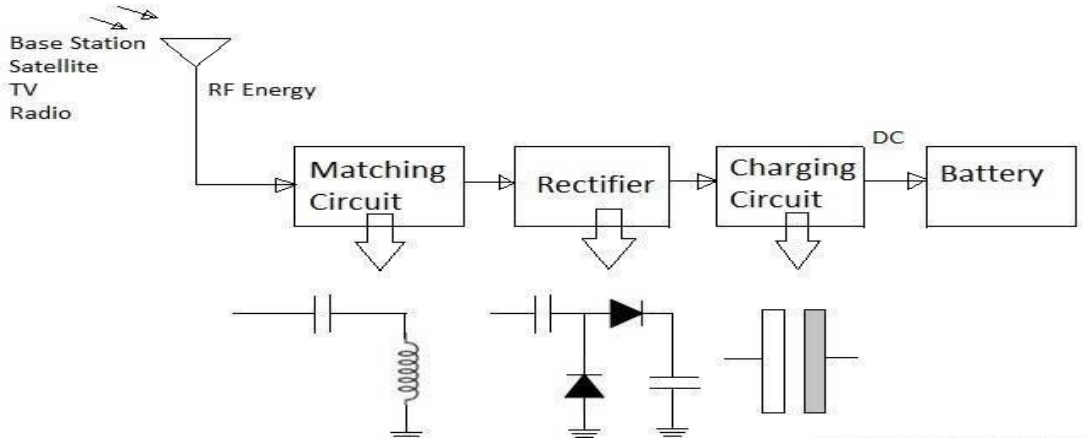


FIGURE: 3.2.3.2.3 RF Harvesting Section

RF Antenna & RF Rectifier Circuit

Captures ambient RF energy (radio waves).

Converts RF AC signals into DC, typically uses diode-based rectification.

Filter Capacitors

Smoothens the rectified signal.

Role: Harvests low-power RF signals and converts them into usable DC energy.

The harvesting paths include:

Solar Panel →Blocking Diode → DC-DC Converter → Storage Bus

Inductive Coil → Bridge Rectifier → Filter Capacitor → Regulator → Storage Bus

RF Antenna → Matching Network → RF Rectifier → Filter → Storage Bus All energy sources converge at a regulated storage node.

3.2.3.2.4 Power Management and Storage Subsystem

This subsystem ensures safe storage and regulation of harvested energy **(Kumar, 2023).**

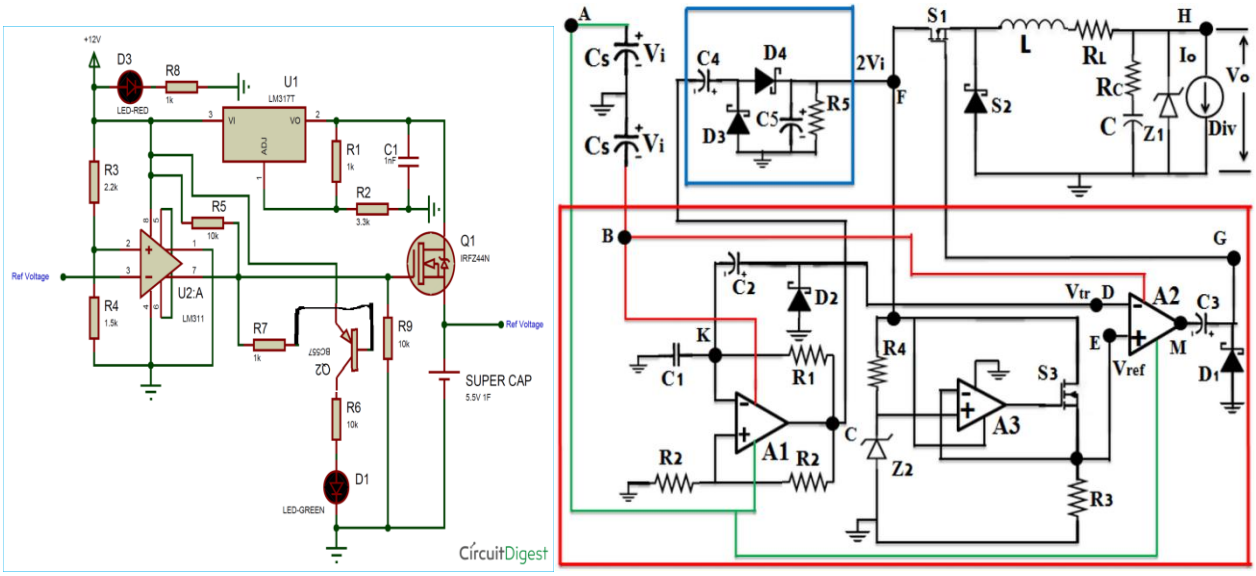


Figure 3.2.3.2.4: Energy Storage Circuit

At this stage there is energy rectification, voltage regulation use of super capacitor to bank voltage and a is divider connected to MCU ADC

The super capacitor is selected due to:

1. High cycle life
2. Fast charge capability
3. Suitability for intermittent power
4. Reduced maintenance compared to batteries

The capacitor voltage is continuously monitored to determine energy state.

3.2.3.2.5 Central Power Management Unit

Function:

Combines outputs from solar, magnetic, and RF harvesters, regulates voltage (VREG), prevents reverse current flow and protects against overvoltage.

Role: Acts as the energy traffic controller of the entire system.

3.2.3.2.5.1 Super capacitor Energy Storage

Stores harvested energy, provides burst current capability for microcontroller operation and handles fluctuations from intermittent harvesting sources.

Role: Acts as the system battery (but faster charge/discharge).

3.2.3.2.5.2 Voltage Divider + ADC Monitoring

Voltage Divider

Scales down super capacitor voltage to safe ADC levels.

ADC (inside Microcontroller)

Measures stored energy voltage, enables energy-aware operation (sleep, wake, duty cycling).

Role: Allows intelligent energy management.

3.2.3.2.6 Energy Storage Calculation

Super capacitor Energy Analysis

The energy stored in a capacitor is calculated using the formula:

$$\mathbf{E = \frac{1}{2} cv^2}$$

Where:
C = 30 F
V = 3.0 V

$$E = 0.5 \times 30 \times (3)^2$$

$$E = \mathbf{135\ Joules}$$

If the system consumes 30 mW average:

$$\text{Runtime} = \frac{\text{Energy}}{\text{Power}}$$

$$\text{Runtime} = \frac{135}{0.03} = 4500 \text{ seconds}$$

$$\text{Runtime} = \mathbf{75\ minutes}$$

This demonstrates that the system can continue operating for over one hour without energy input, improving resilience to intermittent harvesting **(Kularatna, 2011).**

3.2.3.2.7 Processing and Sensing Subsystem

This subsystem performs sensing, adaptive control, and communication.

Components: Ultra-low-power microcontroller, environmental sensors (e.g., temperature, humidity), ADC channel for voltage monitoring and a USB interface for serial transmission.

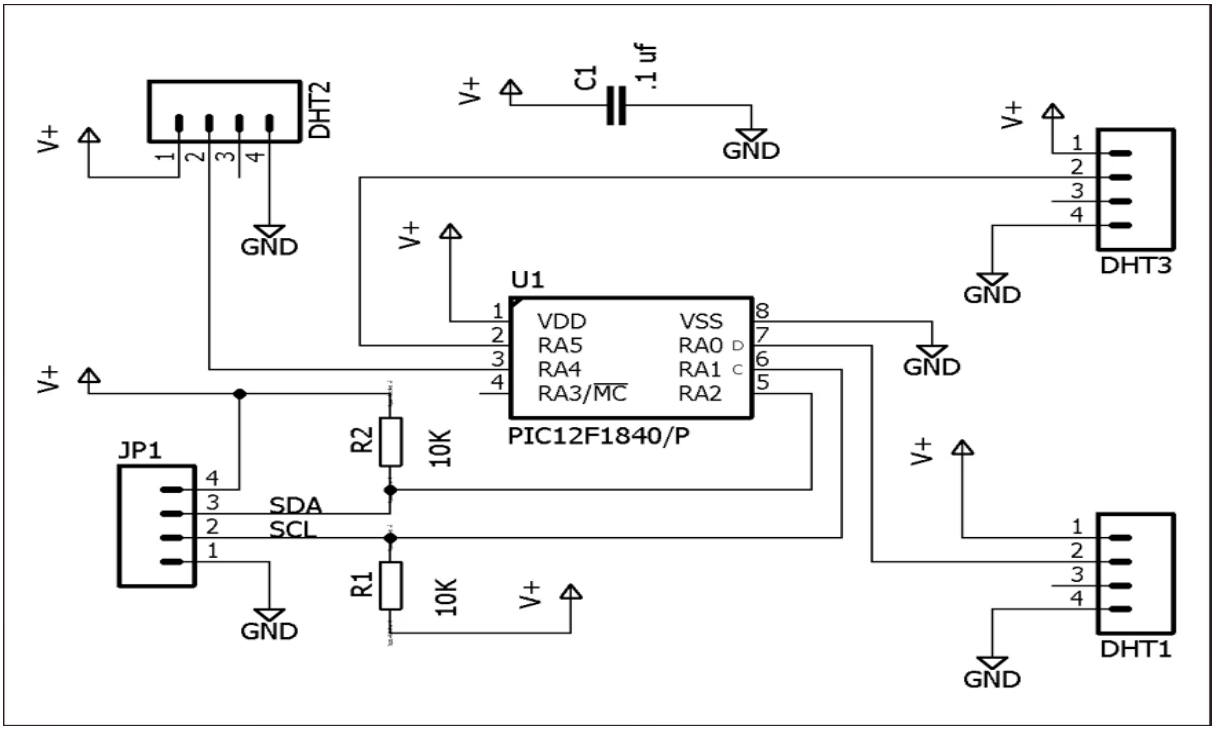


Figure 3.2.3.2.7: MCU and Sensor Interfacing

Connections include:

1. I2C for digital sensors
2. ADC for capacitor voltage measurement
3. GPIO for control signals
4. Regulated Voltage current supply from storage

3.2.3.2.7.1 Microcontroller & Sensors

Microcontroller: It's the core processing unit that reads sensor data, monitors stored energy, controls system operation and manages UART communication.

May communicate via I²C, SPI, or analog input and Powered by regulated 3.3V or 5V from power management.

Role: This is the brain of the system.

Sensors:

1. Temperature sensor.
2. Humidity or environmental sensor.

Power Budget Analysis

$$\mathbf{p = v \times i}$$

$$p = 3.3 \times 0.0195 \approx 64mW$$

If active time = 1 second every 10 seconds average current:

$$I_{avg} = \frac{(19.5 \times 1) + (0.005 \times 9)}{10}$$

$$\mathbf{I_{avg=2mA}}$$

$$\mathbf{P_{avg=6.6mW}}$$

Estimated component current consumption:

Component	Active Current	Sleep Current
MCU	10 mA	5 µA
Temperature Sensor	1.5 mA	2 µA
USB-UART	8 mA	1 mA
Total (Active)	~19.5 mA	—

At 3.3 V: This demonstrates that duty cycling significantly reduces energy consumption **(Kansal et al., 2007).**

3.2.3.2.8 Serial Communication and Data Logging Subsystem

To minimize complexity, the system uses USB serial communication instead of wireless transmission.

Signal flow: Microcontroller UART → USB-UART Converter → USB Cable → Laptop → Python Logger → CSV File

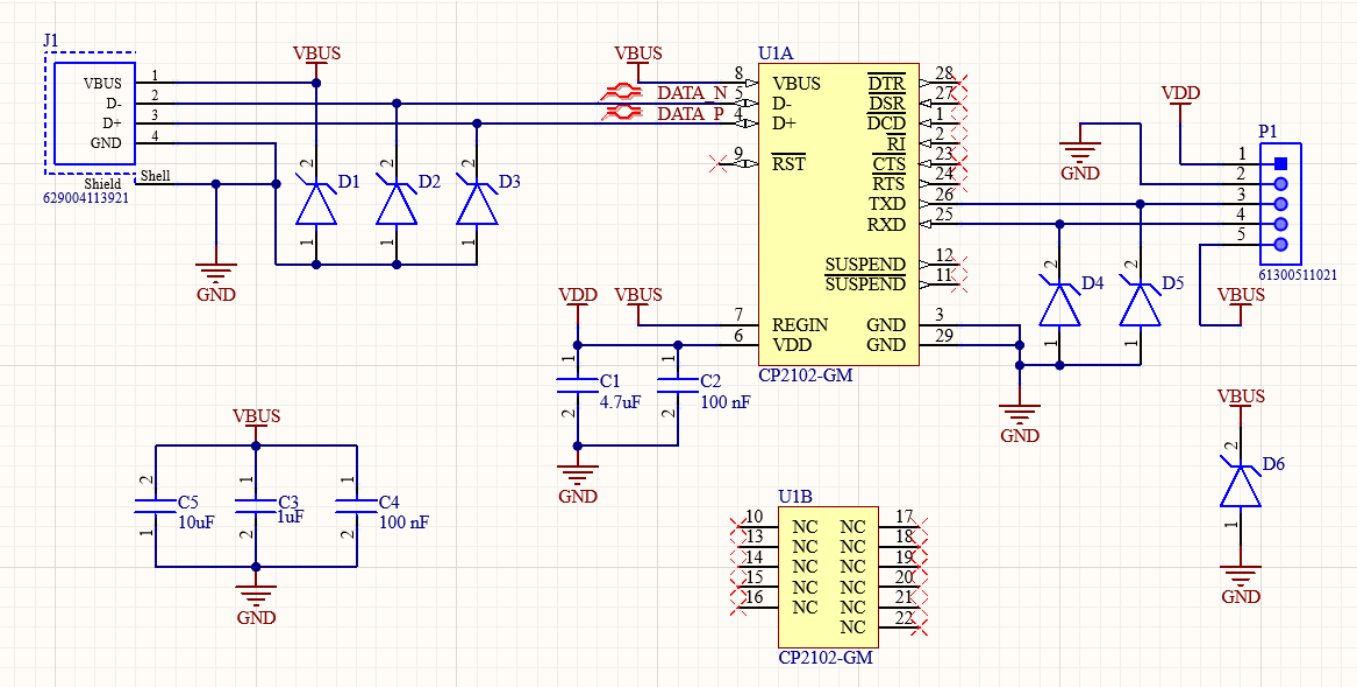


FIGURE: 3.2.3.2.8 Serial Communication and Data Logging Subsystem

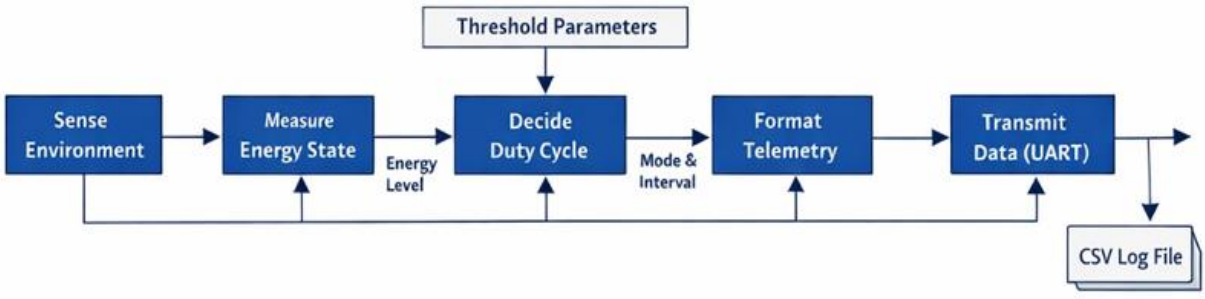


Figure 3.2.3.2.8: Data flow Diagram

3.2.3.2.8 USB-UART INTERFACE

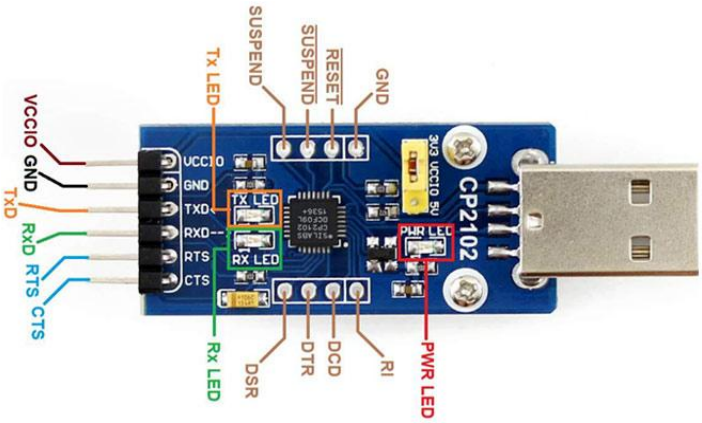


Figure 3.2.3.2.8: USB Serial Interface

Converts MCU UART (TX/RX) signals to USB protocol then allows connection to the laptop in order to enable CSV logging and If USB was removed Data may instead be sent via direct UART pins.

Overall Energy & Signal Flow

Power Path:

1. Solar / Magnetic / RF
   → Rectification
   → Regulation
   → Super capacitor
   → Voltage Regulator
   → Microcontroller + Sensors

Data Path:

2. Sensors
   → Microcontroller
   → UART (TX/RX)
   → Logging system

3.2.3.2.8.1 System Behavior Summary

Component	Primary Role
Solar Panel	Main energy source
Magnetic Harvester	Secondary energy source
RF Harvester	Backup ultra-low energy source
Rectifier	AC → DC conversion
Power Management	Voltage control & merging
Super capacitor	Energy storage
Voltage Divider	Safe voltage sensing
ADC	Energy monitoring
Microcontroller	Processing & control
Sensors	Environmental data collection
UART	Data communication

3.2.4 Stage Four: Firmware Development and Energy-Aware Control Algorithms

At this stage, firmware for the autonomous sensor node is being developed. Embedded software is being designed to monitor energy availability, manage power states, and execute adaptive duty-cycling algorithms. The firmware is dynamically adjusting sensing, processing, and communication operations based on energy levels stored in the super capacitor. Energy-aware scheduling techniques are being implemented to minimize power consumption, prevent brown-out conditions, and maintain stable system operation under fluctuating ambient energy supply.

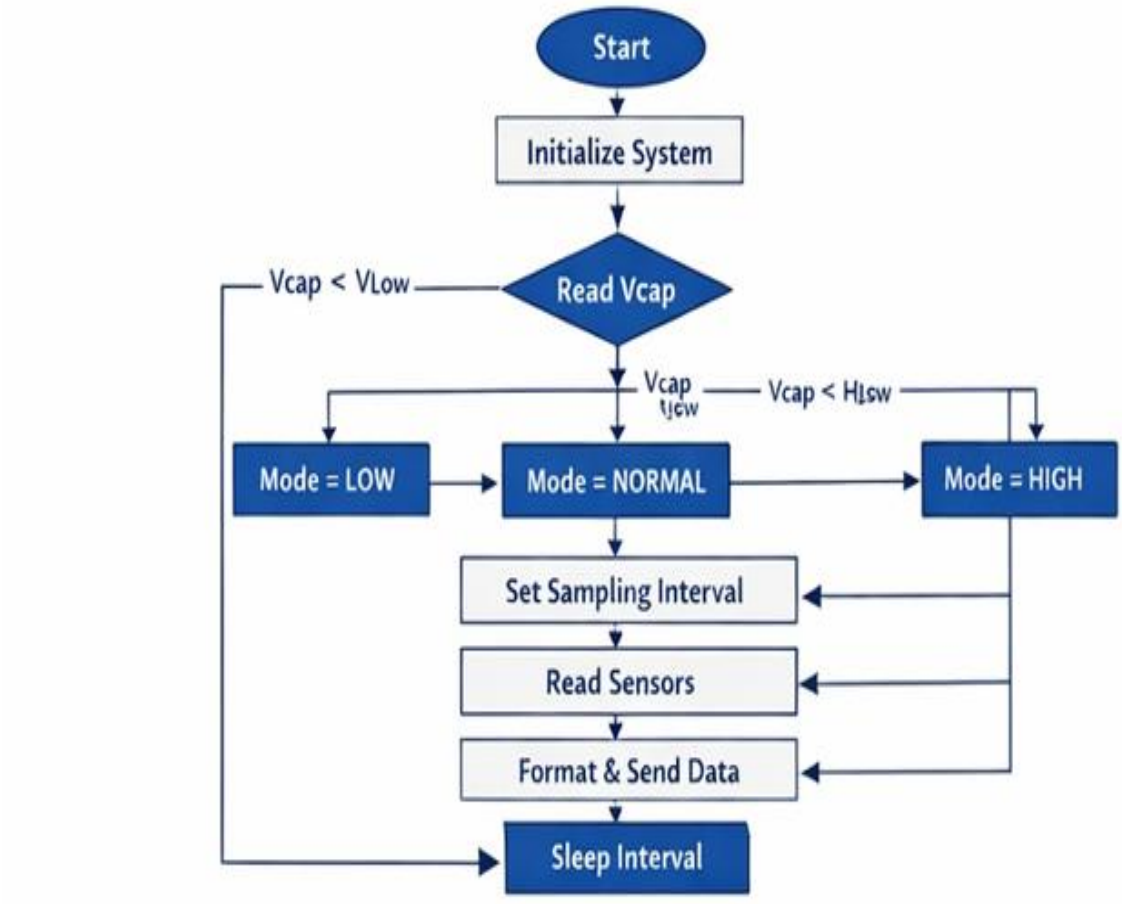


Figure 3.2.4: Firmware operational flowchart for Energy-Aware Duty Cycling

3.2.4.1 Software Architecture

The firmware is organized into the following modules:

3.2.4.1.1 Energy Monitoring Module:

Reads capacitor voltage, converts ADC values to millivolts and determines energy mode based on thresholds.

3.2.4.1.2 Adaptive Duty Cycling Module:

Energy states are defined as:

1. **LOW Mode:** Minimal sensing, extended sleep.
2. **NORMAL Mode:** Standard operation.
3. **HIGH Mode:** Increased sampling frequency.

This ensures energy consumption does not exceed harvested energy.

3.2.4.1.3 Sensor Management Module:

Initializes sensors, acquires readings and powers down peripherals when inactive.

3.2.4.1.4 Serial Transmission Module

Formats telemetry as CSV string, transmits one line per cycle and Maintains stable baud rate communication.

3.2.4.2 Telemetry Data Format

Each transmission follows:

Device_id, Sequence number, Temperature, Humidity, Vc – mv and Energy mode

Example:

NODE01,1052,26.4,57.8,3185, NORMAL

This format ensures compatibility with spreadsheet analysis tools.

3.2.4.3 Duty Cycle Timing Table

Mode	**Sampling Interval**	**Active Time**	**Sleep Time**
LOW	60 s	0.5 s	59.5 s
NORMAL	10 s	1 s	9 s
HIGH	2 s	1 s	1 s

This ensures harvested energy is not depleted faster than replenished **(Mazunga, 2021).**

3.2.4.4 Laptop Logging Application

A Python-based logger performs: Serial port initialization by sending data synchronously through a continuous data capture of Line parsing then CSV file are written and later Timestamp additions.

The CSV structure includes:

| pc_timestamp | device_id | sequence | temperature | humidity | Vc - mv | energy mode |

This enables performance evaluation and graphical analysis.

3.2.4.5 System Operation Flow

The system operates in the following loop:

1. Wake from sleep.
2. Measure capacitor voltage.
3. Determine energy mode.
4. Acquire sensor data.
5. Format telemetry string.
6. Transmit via USB serial.
7. Enter sleep according to selected duty cycle.

This cycle repeats continuously while energy is available.

3.2.5 Stage Five: System Integration, Testing, and Evaluation

In the final stage, all hardware and firmware components are being integrated into a complete functional prototype. The system is being tested under varying environmental and energy availability conditions to evaluate overall performance. Key performance metrics, including energy harvesting efficiency, system uptime, operational reliability, and autonomous functionality, are being measured and analyzed. The experimental results are being compared against the defined system requirements to validate the effectiveness of the proposed multi-source energy-harvesting autonomous sensor node.

Advantages of the Selected Architecture

The chosen architecture provides:

1. Minimal implementation complexity
2. Reduced firmware size
3. No wireless stack overhead
4. No internet dependency
5. Deterministic and stable testing environment
6. Easy debugging
7. Reliable demonstration platform

Most importantly, it ensures the project focuses on energy harvesting performance and adaptive workload control.

Disadvantages of the Selected Architecture

1. Requires physical USB connection.
2. Not suitable for remote deployment.
3. Does not support real-time cloud monitoring.

However, wireless or cloud integration can be implemented as future work without altering the core energy-aware firmware.

Failure Modes and Mitigation

Risk Analysis

Failure Scenario	**Impact**	**Mitigation**
Low solar input	Reduced energy	Multi-source harvesting
Capacitor leakage	Reduced storage	Periodic testing
Overvoltage	Component damage	Regulation stage
USB disconnection	Data loss	Buffered transmission
Brownout reset	MCU reboot	Energy-aware control

This is to improve system robustness and reliability.

3.3 Tools and Technologies to Be Used

In this study, a range of hardware and software tools and technologies such as Arduino IDE are to be utilized to support the design, implementation, and evaluation of the proposed multi-source energy-harvesting autonomous sensor node. Low-power microcontrollers are being employed to execute sensing, control, and communication tasked by use of (usb-uart), while dedicated energy harvesting power management integrated circuits (PMICs) are being used to efficiently regulate and store energy harvested from ambient solar, magnetic, and radio frequency sources. Super capacitor-based energy storage technologies are being implemented to buffer harvested energy and ensure stable system operation under intermittent energy conditions.

4. CONCLUSION

This chapter presented the complete system design of the Multi-Source Energy-Harvesting Autonomous Sensor Node using a minimal USB serial data logging architecture.

By eliminating wireless and cloud complexity, the system achieves:

Reduced implementation risk, Improved stability, Lower energy overhead and Clear focus on adaptive energy management.

The architecture provides a reliable and reproducible platform for experimental evaluation and performance validation.

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